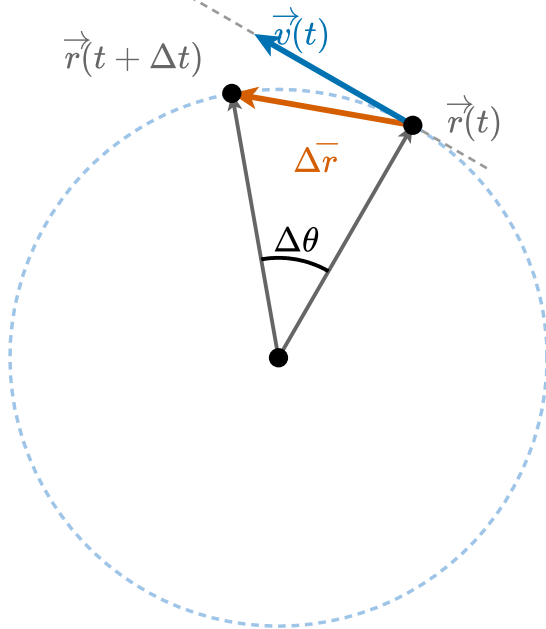


tangent line



$$\|\Delta \vec{r}\| = 2r \sin \frac{\Delta \theta}{2} \approx r \Delta \theta$$

$$\Rightarrow v = \lim_{\Delta t \rightarrow 0} \frac{\|\Delta \vec{r}\|}{\Delta t} = r \omega$$